2. 易知 $\langle Z, *\rangle$ 为单位元 $e=2$ 的群

下证：$\langle Z, +, 0\rangle \cong \langle Z, \times, 2\rangle$

构造 $h: Z \to Z,\ x \longmapsto x+2$

$h(m+n) = m+n+2 = (m+2)+(n+2)-2 = h(m) \times h(n)$

故 h 为同态，只需证 h 为双射.

① 单射：$h(x_1)=h(x_2) \Longrightarrow x_1+2=x_2+2 \Rightarrow x_1=x_2$

② 满射：$\forall y \in Z$，$y=x+2$，$\therefore \exists x_0=y-2 \in Z$. $h(x_0)=y$

4. $G=\{e\}$ 时，只有平凡子群 $\{e\}$.

$G \neq \{e\}$ 时，G 必然为循环群，理由如下：

$\forall a \in G, a \neq e$，$\therefore \langle a\rangle = \{a, a^2, a^3, \dots, e\}$

又 $\langle a\rangle \in G$，又 G 无非平凡子群，$\therefore \langle a\rangle = G$，

$\therefore$ G 可由任一非单位元生成，$\therefore$ G 为循环群.

下证 $|G|=p=|a|$ 为质数：假设 $\exists m \in (1, p)$，$m \mid p$，

令 $k=\frac{p}{m}$，$\therefore H=\{a^m, a^{2m}, \dots, a^{(k-1)m}, e\} \leq G$

$|H| = |a^m| = \frac{p}{(p,m)} = \frac{p}{m} = k \in (1, p)$.

$\therefore$ H 为 G 的非平凡子群. 矛盾. $\therefore$ $|G|$ 为质数

6. $\exists h: G \to G'$，$a \longmapsto h(a)$ 为满射

设 $G=\langle a\rangle = \{a^n \mid n \in Z\}$

$G' = h(G) = h(\langle a\rangle) = h(\{a^n \mid n \in Z\})$

$= \{[h(a)]^n \mid n \in Z\} = \langle h(a)\rangle$.

$\therefore$ G' 也为循环群.

8. $C_m \cong \langle N_m, +_m, 0\rangle$

$C_m \times C_n = \{\langle a, b\rangle \mid a \in C_m, b \in C_n\}$

构造 $h: N_{mn} \to N_m \times N_n$，$h(1) = \langle 1, 1\rangle$

$h(a) = h(1 +_{mn} 1 +_{mn} \cdots +_{mn} 1) = h(1) \circ h(1) \circ \cdots \circ h(1)$

$= \langle 1,1\rangle \circ \langle 1,1\rangle \circ \cdots \langle 1,1\rangle = \langle a \bmod m, a \bmod n\rangle$

$\therefore h(N_{mn}) = \{\langle a \bmod m, a \bmod n\rangle \mid 0 \leq a < mn-1\}$

下证：$\gcd(m,n)=1 \iff h$ 为双射

① 先证 $\gcd(m,n)=1 \Longrightarrow h$ 为双射

(i) 单射：最小公倍数 $\mathrm{lcm}(m,n) = \frac{mn}{\gcd(m,n)}$

$\forall a, b \in N_{mn}$，$h(a)=h(b) \Rightarrow \begin{cases} a \bmod m = b \bmod m \\ a \bmod n = b \bmod n \end{cases}$

$\therefore (a-b) \bmod m = (a-b) \bmod n = 0$

$\therefore a-b = k \cdot \mathrm{lcm}(m,n)$，$k \in Z_+$，又 $a, b \in [0, mn)$

$\therefore a-b=0$，$\therefore a=b$，则 h 为单射.

(ii) 满射：$|N_{mn}| = mn = |N_m \times N_n|$. 又 h 为单射，

$\therefore$ h 为满射.

综上，h 为双射.

② 再证 h 为双射 $\Longrightarrow \gcd(m,n)=1$

假设 $\gcd(m,n)=d>1$，$\therefore \mathrm{lcm}(m,n) = \frac{mn}{d} < mn$

取 $a = \mathrm{lcm}(m,n) \in [0, mn)$. $h(a) = \langle 0, 0\rangle = h(0)$.

$\therefore$ h 不为双射，矛盾

综上：m, n 互质 iff h 为双射，$\therefore N_{mn} \cong N_m \times N_n$

$\therefore$ $C_{mn} \cong C_m \times C_n$

1. $H = \{e, a^4, a^8\} = Ha^4 = Ha^8$

$Ha = \{a, a^5, a^9\} = Ha^5 = Ha^9$

$Ha^2 = \{a^2, a^6, a^{10}\} = Ha^6 = Ha^{10}$

$Ha^3 = \{a^3, a^7, a^{11}\} = Ha^7 = Ha^{11}$

2. $H = \left\{ \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ 1 & k_2 & k_3 & \cdots & k_n \end{pmatrix} \middle| \begin{pmatrix} 2 & 3 & \cdots & n \\ k_2 & k_3 & \cdots & k_n \end{pmatrix} \text{为} (2,3,\dots,n) \text{上任意置换} \right\}$

H 在 S_n 中所有左陪集的集合为：

$\left\{ \left\{ \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ k & k_2 & k_3 & \cdots & n_k \end{pmatrix} \middle| \begin{pmatrix} 2 & 3 & \cdots & n \\ k_2 & k_3 & \cdots & n_k \end{pmatrix} \text{是} \{2,3,\dots,n\} \text{到} \{1,2,3,\dots,n\}-\{k\} \text{上的双射} \right\} \middle| k=1,2,3,\dots,n \right\}$

4. $\forall a \in H \cap K$，$a \in H$，$a \in K$

由定理：$|a| \mid |H|$，$|a| \mid |K|$，又 $\gcd(|H|, |K|) = 1$

$\therefore |a|=1$，$\therefore a=e$. 故 $H \cap K = \{e\}$